

# Application Code & Replication Benchmark Suite

Technical Documentation & MATLAB–R Bridge Setup Guide

Replication Package for:

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Authors:

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## 1 Overview & Repository Structure

This repository contains every single routine and the exact datasets required to replicate all empirical applications and simulations presented in the manuscript.

All source code files (.m and .R) are self-documented. Detailed mathematical definitions, input/output specifications, diagnostic algorithms, and option flags are located directly within the header comment blocks of each individual script.

## 2 MATLAB–R Bridge Setup Guide

The MATLAB pipeline delegates specialized statistical operations to R using dedicated bridge scripts:

- **Polychoric Estimation** (`srot.R`): Executed by `processData.m` for ordinal, Likert-scale, or mixed continuous-ordinal data using the `psych` package.
- **Oblique Factor Rotations** (`calc_rotation.R`): Executed for specialized oblique gradient projection rotations (e.g., Bentler, Geomin, Infomax, McCammon) using the `GPArotation` package.

Follow the steps below to configure the local bridge environment on your system.

### Operating System Compatibility Note:

The automatic detection of R's binary path is optimized for **Windows** systems. macOS and Linux users can execute the bridge seamlessly by ensuring that `Rscript` is available in their system's global `PATH`.

### 2.1 Prerequisites & Dependencies

1. **MATLAB**: R2018b or newer recommended.
2. **R Environment**: Download and install the latest version from CRAN (<https://cran.r-project.org/>).
3. **Required R Packages**: Open R or RStudio and run:

```
install.packages(c("psych", "GPArotation", "R.matlab"))
```

## 2.2 Directory Architecture Requirements

The pipeline uses a fixed file exchange directory inside your system's **Documents** directory to exchange data frames and matrix operations between MATLAB and R.

### 1. Locate system Documents directory:

- **Windows:** C:\Users\\Documents\
- **macOS:** /Users/<YourUsername>/Documents/
- **Linux:** /home/<YourUsername>/Documents/

### 2. Position the exchange folder: Copy or place the entire matlab\_bridge folder directly from the GitHub repository into the rfiles/ subfolder inside your Documents directory:

```
.../Documents/rfiles/matlab_bridge/
```

### 3. Verify script locations: Ensure both R bridge functions are located inside that exchange folder:

- .../Documents/rfiles/matlab\_bridge/calc\_polychoric.R
- .../Documents/rfiles/matlab\_bridge/srot.R

## 2.3 Execution Workflows

### 2.3.1 1. Polychoric & Mixed Correlation Estimation

When calling `processData(X, Q, opts)` in MATLAB with `opts.is_ordinal = 1`, the system executes the following sequence:

```
[MATLAB] Exports matrix X --> temp_X_data.mat
      |
      v
[Rscript Executed via System Call]
calc_polychoric.R (psych::polychoric / mixedCor)
      |
      v
[MATLAB] Imports matrix R <-- temp_R_poly.mat
```

### 2.3.2 2. Specialized Oblique Factor Rotations

When executing factor rotations requiring Gradient Projection Algorithms via R, the system executes:

```
[MATLAB] Exports matrix A_norm --> temp_A_norm.mat
      |
      v
[Rscript Executed via System Call]
srot.R (GPArotation)
      |
      v
[MATLAB] Imports Lambda & T <-- temp_rot_out.mat
```

### 3 Troubleshooting & Fallback Mechanisms

- **Custom R Paths (Windows):** If R is installed outside default paths (C:\Program Files\R\), add R's bin directory to your system's PATH environment variable.
- **Terminal Test (macOS/Linux):** Verify accessibility by entering `Rscript --version` in your terminal.
- **Automatic Fallback Guardrails:**
  - If the correlation bridge fails or does not converge, the pipeline generates a warning and falls back to standard Pearson pairwise correlations.
  - If the rotation bridge fails, MATLAB falls back to internal orthogonal/oblique rotation solvers.